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Research papers

Mapping of small water bodies with integrated spatial information for time series images of optical remote sensing

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ABSTRACT

Small water bodies and their temporal changes are, especially in urban areas, closely related to people's daily life and they have an impact on the living environment. For small water bodies mapping with optical remote sensing images, it is challenging to establish a balance between reducing incorrect water detection and increasing the integrity of water extraction. For time-series application, the temporal variability of the spectral information is also challenging for the widely used threshold methods, which are frequently solely based on spectral analysis. In this work we propose a spatial information-integrated small water bodies mapping (SWM) method to achieve a complete and accurate extraction and temporal change monitoring of small water bodies in complex urban environments. Our strategy is to make use of the spatial contextual information to account for the indistinguishability of small water bodies in spectral information. The roughness of the water index is calculated to enhance the contrast between water bodies and other thematic classes eventually present in the imagery. The proposed SWM was applied to different water indexes with an automatic threshold determination. We tested the effectiveness of the proposed algorithm using Landsat and Sentinel-2 multispectral data from three different urban environments (Shanghai, Guangzhou, and Wuhan in China) which include a variety of river courses and lakes. Nantan Lake of China is selected as a representative experimental area to test the SWM method by generating the inter- and intra-year water results. Overall accuracy (OA), F1 score (F1), producer's accuracy (PA), and user's accuracy (UA) were used to quantitatively evaluate the accuracy of the algorithm. Compared with the four state-of-the-art water detection methods (supervised random forest classification, hierarchical clustering, multi-band threshold, and modified normalized difference water index (MNDWI)), the proposed SWM algorithm achieves better water extraction performance. Small water bodies are found to be extracted more completely and incorrect water extractions are alleviated. The overall accuracy of the SWM algorithm achieves an average of approx. 97% (OA) and 0.95 (F1). The long-time sequence (from 2007 to 2021) and the short-time interval (monthly results of the year 2020) of the water extraction results by the SWM agree well with the ground truth data. The mean absolute area deviations between the water body extraction results of the SWM algorithm and ground truth is significantly smaller than that of the Global Surface Water dataset developed by the European Commission's Joint Research Centre (3.5% vs 49.0% for annual results and 6.0% vs 35.0% for monthly results).

1. Introduction

Small water bodies, including ponds and small lakes, low-order streams, ditches and springs, are closely related to people's daily life and have an impact on the living environment (Biggs et al., 2017; Ogilvie et al., 2018a; Riley et al., 2018). Given the changes in land use/cover and various environmental changes in the process of urbanization, surface water including small water bodies is undergoing temporal and spatial changes. These may lead to negative effects concerning the urban climate or recreation, or even to disasters that endanger the safety of life

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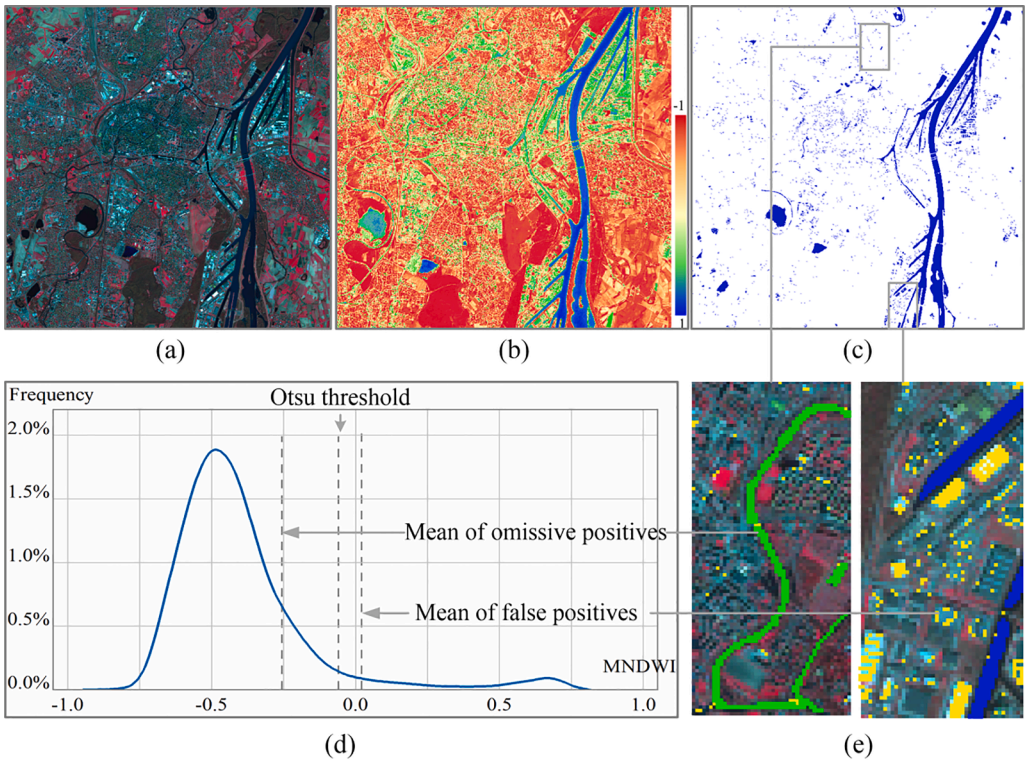


Fig. 1. An example of surface water extracted using the water index at Strasbourg in France. (a) The false-color map of Sentinel-2 data. (b) The corresponding MNDWI map. (c) The result of the water bodies extracted using the Otsu threshold. (d) The histogram of the water index. (e) The close-up of some characteristic areas in the water extraction result. The blue label is the correct extraction result. The yellow label is the false positives (incorrect water extractions) and the green label represents the false negatives (undetected water bodies). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and property, such as floods and water shortages (Chawla et al., 2020; Ogilvie et al., 2018b; Zou et al., 2017; Taubenböck et al., 2011). Therefore, monitoring small water bodies for a better understanding of its local-specific effects is essential for better informed urban planning decisions (Carstens and Amer 2019; Huang et al., 2015).

Small water bodies with limited size and large numbers pose significant challenges for high-precision mapping. Optical remote sensing provides an effective data source to map these small surface water bodies. With the availability of the Landsat and Sentinel-2 multispectral data, rapid and long-term monitoring of surface water have been widely applied in water resource monitoring (Meng et al., 2020), flood detection and evaluation (DeVries et al., 2020), wetland mapping (Mao et al., 2020), and lake change detection (Gu et al., 2021).

The surface water extraction algorithms for optical images can be mainly classified into two kinds: image classification and spectral analysis (Chen et al., 2020c). Supervised image classification with machine learning algorithms is an effective and direct way to extract surface water using optical images (Pickens et al., 2020; Wu et al., 2021), such as SVM (Sun et al., 2015), random forest (Ko et al., 2015) and deep learning (Isikdogan et al., 2017; Wang et al., 2020). These supervised classification methods depend on the quality and amount of the training samples, and can achieve satisfying water extraction results under the premise that there are sufficient representative samples. In addition, some ancillary data, such as a Digital Elevation Model (DEM) is supporting optical images to obtain satisfactory water extraction results (Pekel et al., 2016), which limits the extensive use of image classification method in the mass processing of remote sensing data (Ogilvie et al., 2020).

In contrary to this image classification approach, the method based on spectral analysis does not rely on training samples and is realized by thresholding the water index image which is established to enhance the contrast between water bodies and other land covers (Li et al., 2021; Lu et al., 2020). Various water indexes have been proposed based on spectral information and the most commonly used water indexes include the normalized difference water index (NDWI) (McFeeters 1996), the modified normalized difference water index (MNDWI) (Xu 2006), and the automated water extraction index (AWEI) (Feyisa et al., 2014). The threshold can be set empirically (Feyisa et al. 2014) (e.g. 0) or automatically. The currently widely used automatic threshold determination method is Otsu (1979) by maximizing the variance between water bodies and other land covers (Cordeiro et al., 2021; Pan et al., 2020).

The water index approaches by spectral analysis mainly rely on selecting appropriate thresholds to classify water bodies and non-water bodies. However, the variability of spectral information brings challenges to setting appropriate transferable thresholds as settings are often specific to each study case (Campos et al., 2012; Klein et al., 2017). Specifically, some low-albedo targets (such as shadows) and high-albedo targets (such as clouds, built-up areas, etc.) often have a water index value similar to or higher than the water body, which may create several false positives (incorrect extractions), especially in complex urban scenes (Fisher et al., 2016; Yang et al., 2018). To reduce false positives, the threshold to segment the water index map may need to be set relatively higher (closer to value 1). On the other hand, small water bodies tend to have relatively small water index values compared to pure water bodies, due to the influence of the surrounding environment and mixed pixels. To completely extract small surface water bodies, it is best to set the corresponding threshold value lower (closer to −1). This dilemma to reduce false positives and increase the extraction integrity of small water bodies by adjusting the threshold is illustrated in Fig. 1. The mean of incorrect water extractions is greater than the Otsu threshold, but at the same time the mean of undetected water bodies is less than the Otsu threshold. It is thus challenging to adjust the threshold to reduce incorrect water extractions and increase the precision of detected water bodies at the same time. In addition, it is especially important and difficult to determine the optimal threshold for time-series analysis of water bodies (Yao et al., 2019). Changes in the state of the water bodies will affect their corresponding spectral albedo. Different atmospheric conditions, angles of incidence of sunlight, or radiometric calibration states of time-series remote sensing images acquired by different sensors will also cause variation in the image radiation values.

To overcome the instability of spectral information and achieve sufficient, accurate observations of long-term surface water dynamics of

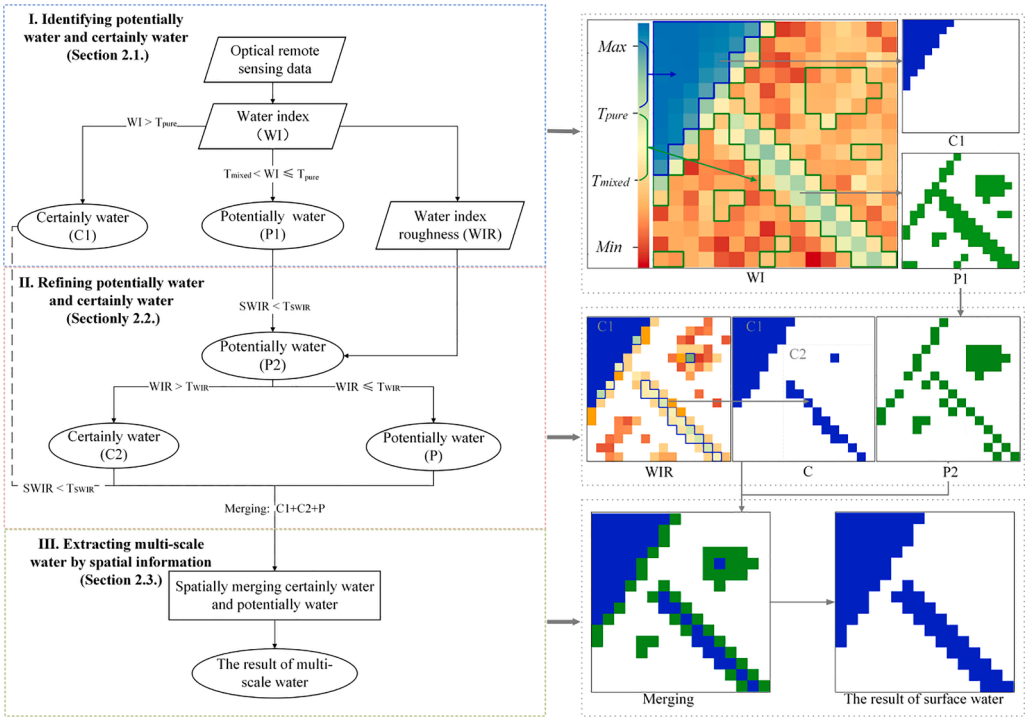


Fig. 2. The framework of the proposed spatial-information-integrated small water bodies mapping algorithm (SWM).

the small water bodies, we propose a spatial information-integrated small water bodies mapping algorithm (SWM) for time-series of Landsat and Sentinel-2 optical remote sensing imagery. In the SWM algorithm, the spatial contextual information is exploited to compensate for the uncertainty of spectral information and to refine and extract small water bodies from multiple perspectives of feature mining and decision-making. The main contributions and works can be summarized as follows:

(1) To avoid the omission of small water bodies, we divide the surface water bodies hierarchy into three probability parts ('certainly water', 'potentially water', and 'certainly non-water') based on spectral analysis. Although the spectral information of small water bodies varies greatly, they can be included in the categories of 'certainly water' or 'potentially water' by a significantly lower threshold.
(2) To refine small water bodies from 'potentially water', we develop a water index roughness feature based on local spatial information to measure spatial variability of surface water. Small water bodies will be more refined from 'potentially water' to 'certainly water' categories, due to the large spatial variability of small water bodies and surrounding land cover categories.
(3) To accurately extract small water bodies, we analyze the spatial relationship of these identified 'potentially water' and 'certainly water' categories to eliminate the interference from other non-water land cover types. More complete small water bodies can be extracted by searching for pixels spatially connected to 'certainly water' from 'potentially water'.
(4) To verify the proposed SWM algorithm, we test its performance in terms of large spatial coverage (a complete scene) and long time-series dynamics at a region. We test whether small water bodies are extracted more complete and incorrect water extractions are alleviated in comparison with several state-of-the-art water extraction methods and products, such as the Global Surface Water dataset developed by the European Commission's Joint Research Centre (hereafter JRC product).

2. Methodology

To obtain the spatial distribution of small water bodies using optical remote sensing imagery, a spatial information-integrated small water bodies mapping algorithm (SWM) technique is proposed. As shown in Fig. 2, the SWM algorithm consists of the following three steps:

In the first step, the probabilistic categories 'certainly water surfaces' and 'potentially water surfaces' are identified by dividing the water body index hierarchically. To reduce the false positives and improve the accuracy of small water bodies, the water index roughness and the SWIR band are used to refine both categories by analyzing the 'potentially water' pixels in the initial step. The obtained classes 'certainly water' and 'potentially water' are spatially merged to achieve accurate surface water. The details of the proposed SWM algorithm are given in the following three sections.

2.1. Identifying the categories 'potentially water' and 'certainly water'

The water index is usually used to extract surface water by setting a threshold since it can increase the difference between water bodies and non-water objects by combining multiple bands of optical remote sensing data. However, considering the complexity of environments and the variability of surface water appearances, there may be overestimation or underestimation of small water bodies only using a single threshold. Generally, the histogram of the water index usually shows bimodal distributions: One peak represents water bodies and another peak represents non-water land cover types. These two parts can be easily distinguished by a single threshold. However, the middle part of these two parts in the histogram represents a mixture of water and non-water materials. The traditional threshold selection method usually chooses the threshold at the bottom of the histogram. Therefore, no matter whether the threshold is set to be higher or lower, it is difficult to extract water bodies with satisfactory results. In the proposed SWM method, we consider the part that contains both, water bodies and non-water land cover types, between the two peaks of the histogram as 'potentially water'.

Generally, most pure water pixels have higher water index values

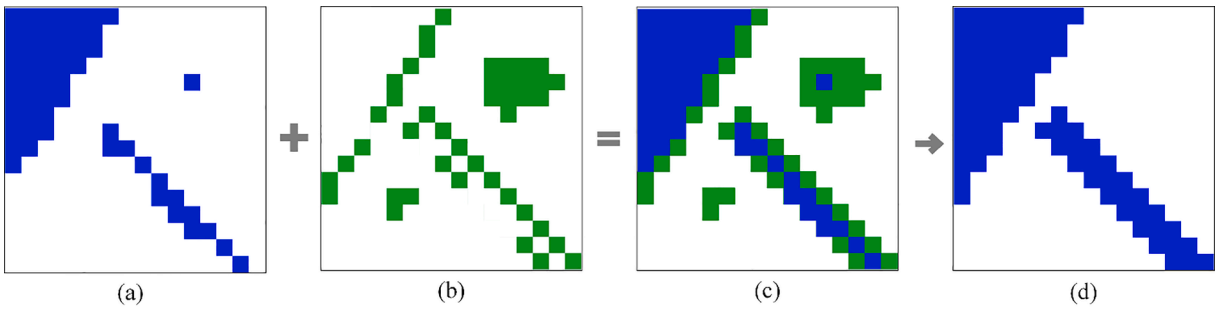


Fig. 3. An example of water extraction based on spatial analysis. (a) 'Certainly water' class. Pixels labeled in blue are 'certainly water'. (b) 'Potentially water' class. Pixels labeled in green are 'potentially water'. (c) Connected segmentation regions. (d) Final water extraction result. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

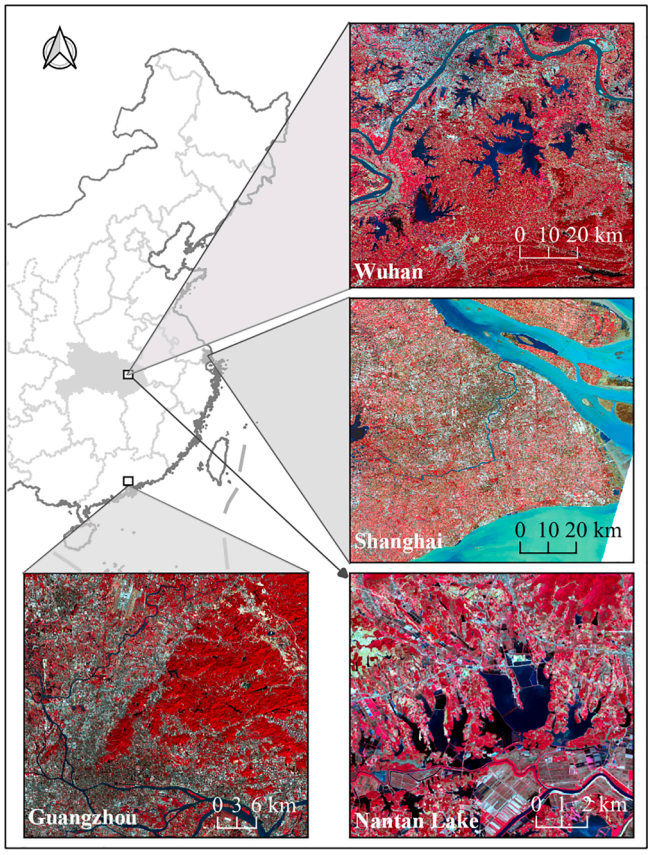


Fig. 4. Study areas and experimental data. The study areas include Guangzhou, Wuhan, Shanghai, and Nantan lake of China. The four false-color images of the Sentinel-2 MSI data show the overview of the study areas.

Table 1
Description of optical images used in this study.

Main study area	Sensors	Acquisition time
Wuhan	Sentinel-2A (MSI)	2019-09-22
Shanghai	Sentinel-2B (MSI)	2018-03-10
Guangzhou	Sentinel-2A (MSI)	2020-10-26
Nantan Lake	Landsat 5 (TM)	2007-08-09
		2011-08-20
	Landsat 7 (ETM+)	2013-08-01
		2020-06-17
		2020-08-04
		2020-09-05
		2020-10-23
		2020-11-08
	Landsat 8 (OLI)	2016-08-01
		2019-07-25
		2020-02-18
		2020-03-05
		2020-08-28
		2021-07-30
	Sentinel-2A (MSI)	2016-07-29
		2020-01-20
		2020-02-09
		2020-04-09
		2020-05-19
		2020-08-17
		2020-11-15
		2021-08-02
	Sentinel-2B (MSI)	2019-07-29
		2020-03-15
		2020-10-11
		2020-12-22

and exhibit relatively large differences to non-water pixels in all water indexes. In this study, we consider these pure water pixels as 'certainly water' (C1), which can be identified based on an automatically selected high threshold (T_{pure}) and is described by Eq. (1):

$$C1 = \{ i \in V | WI_i > T_{pure} \} \quad (1)$$

where V is a set of pixels, and WI_i is the water index value of pixel i. The water index used here and in the method section is representative for the most widely used water indexes including the NDWI, MNDWI, and AWEI (Fisher et al. 2016; Ma et al., 2019). All of these can be used directly, since the proposed method mainly exploits spatial information based on a certain water index to effectively improve the extraction ability of small water bodies. Different water indexes are tested in Section 4.2 to verify the capability for generalization. For the convenience of the description, the common MNDWI is used by default. The high threshold (T_{pure}) can be used to extract this category. However, it is difficult to extract some water bodies of very small sizes, such as small lakes, ponds or narrow rivers, due to the small difference in the water index between them and non-water materials. To extract these water bodies at these small scales, a low threshold (T_{mixed}) is set, and the pixel i will be treated as 'potentially water' (P1) if its water index is greater than T_{mixed} and less than or equal T_{pure}. The 'potentially water' category represents a pixel that potentially shows a water surface, and can be obtained based on Eq. (2).

$$P1 = \{ i \in V | T_{mixed} < WI_i \leqslant T_{pure} \} \quad (2)$$

This category can be included in water bodies at small scales by setting a smaller threshold. Undoubtedly, non-water materials will also be included in the category 'potentially water', which can be effectively removed by spatial information in the following section.

To extract the categories 'certainly water' and 'potentially water', the high threshold (T_{pure}) and the low threshold (T_{mixed}) need to be selected. Considering the bimodal distribution of the water index map,

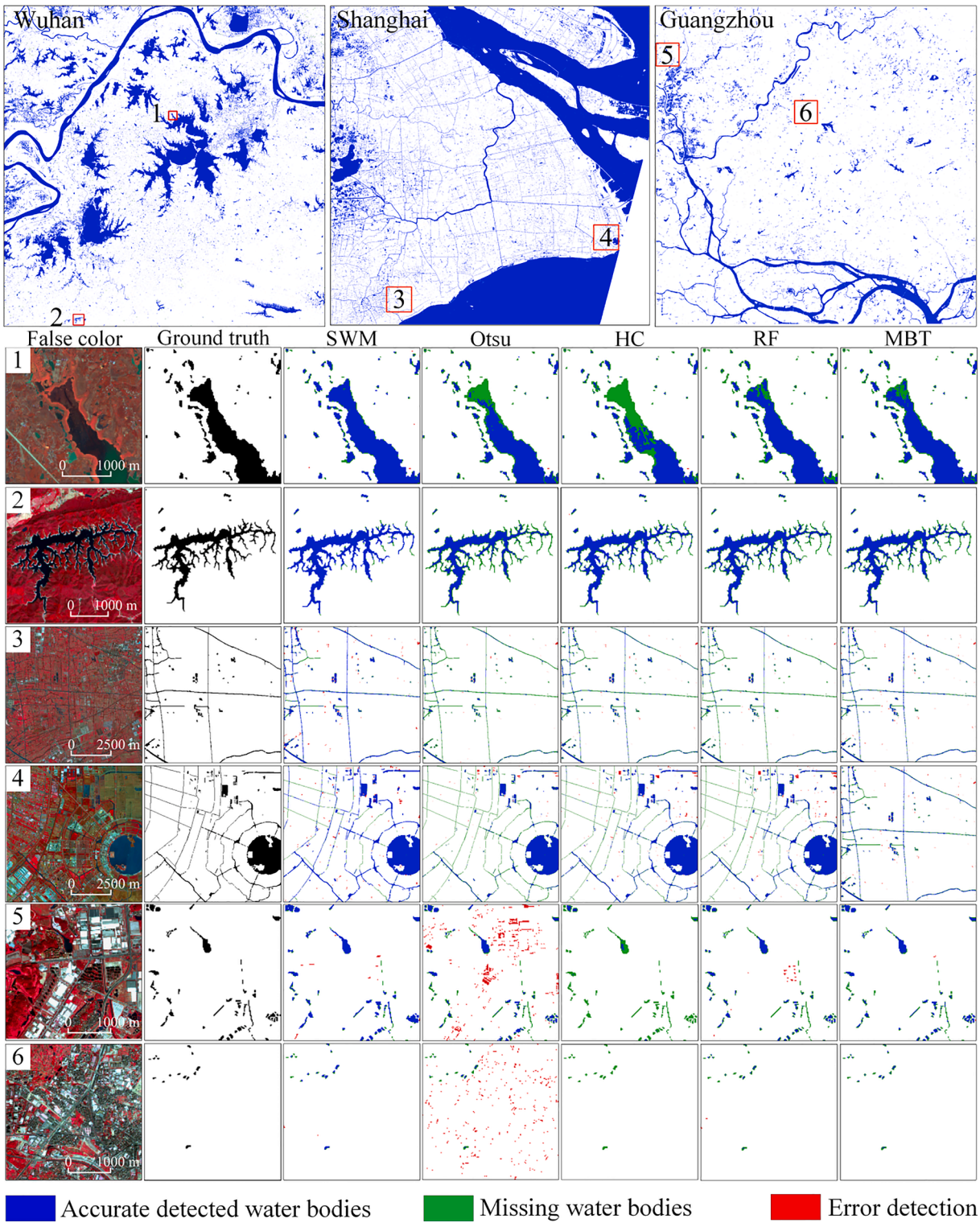


Fig. 5. The water extraction results using the proposed SWM method and comparisons to other algorithms in Wuhan, Shanghai and Guangzhou with six enlarged sample regions.

we propose an automatic thresholding method to select high and low thresholds, which can be illustrated as follows:

$$T_{pure} = \max\{T_s, \quad 0.5 \times (M_1 + std_1)\} \tag{3}$$

$$T_{mixed} = \min\{T_s, \quad 0.5 \times (M_2 - std_2)\} \tag{4}$$

where T_s is the initial threshold of the water index, which can be obtained by experience or by automatic threshold determination by the Otsu approach. Based on the threshold T_s, the water index can be divided into two parts as a whole. One part is a water index greater than the threshold T_s, which contains most of the water bodies. The median and variance of this part of the water index can be calculated and expressed as M_1 and std_1. We combine the median and variance of the water index that contains most of the water bodies to calculate the high threshold, as

Table 2
Water extraction accuracies for all the experimental areas in comparison to different classification methods.

Study area	Method	OA (%)	PA (%)	UA (%)	F1
Shanghai	SWM	**95.87**	**93.50**	94.06	**0.9378**
	Otsu	92.20	77.90	**98.36**	0.8694
	HC	92.87	84.30	93.67	0.8874
	RF	92.43	83.00	93.57	0.8797
	MBT	93.73	87.80	93.01	0.9033
Wuhan	SWM	**97.23**	**93.20**	98.42	**0.9574**
	Otsu	88.53	71.70	92.16	0.8065
	HC	94.23	83.78	98.70	0.9063
	RF	92.57	78.70	98.75	0.8759
	MBT	94.03	83.10	**98.81**	0.9028
Guangzhou	SWM	**97.20**	**98.40**	93.54	**0.9591**
	Otsu	92.53	92.00	86.47	0.8915
	HC	86.73	60.20	**100.00**	0.7516
	RF	94.60	89.60	93.92	0.9171
	MBT	96.53	92.00	97.46	0.9465

shown in Eq. (3). The initial threshold T_s is often obtained by traditional threshold methods (such as experiential or automatic definition of thresholds). To alleviate the influence of non-water materials when obtaining the category 'certainly water', the high threshold T_{pure} will not be less than T_s. On the other hand, the other part is a water index less than or equal to the threshold T_s, and is mainly composed of non-water land cover types. Their corresponding median and variance can be expressed as M_2 and std_2. Thereby, the low threshold T_{mixed} can be calculated by combining T_s, M_2, and std_2 to extract water bodies at all scales except for 'certainly water' as 'potentially water', as shown in Eq.. It is worth noting that if the high threshold T_{pure} and the low threshold T_{mixed} are set to T_s at the same time, the proposed algorithm is equivalent to the traditional water extraction method using experience or automatic approaches for thresholding. Therefore, the traditional water extraction method based on a water index can be regarded as a special case of the proposed method.

2.2. Refining the categories 'potentially water' and 'certainly water'

The obtained categories 'certainly water' (C1) and 'potentially water' (P1) divide the water bodies into two parts based on their probability. These will be merged (cf. Section 3.2) based on the spatial relationship between both categories to achieve an accurate water extraction. However, the high threshold to obtain the category 'certainly water' is often set to be relatively large to alleviate the influence of non-water materials, and the low threshold is often set to be relatively small to include all water bodies with different scales in the category 'potentially water'. This may result in a certain degree of misidentified water bodies and undetected water bodies. On the one hand, some non-water land cover types in areas of 'potentially water' (P1) that are spatially connected to water bodies may be misidentified into water regions that cause incorrect detection, such as bridges, dark roads, and shadows. On the other hand, there are still some independent water bodies that have not been detected, since their water index values are lower than pure water. Therefore, we will refine the 'certainly water' and 'potentially water' categories by increasing the number of the class 'certainly water' and reducing the non-water materials in 'potentially water' P1 based on the SWIR band and the water index roughness in this section.

To prune non-water land cover types in the category 'potentially water', the SWIR band is used; this band has been widely used for water extraction. Water bodies have physically strong absorption characteristics in the SWIR band. The reflectance of surface water in the SWIR band is in general lower than of non-water land cover types (Olthof et al., 2015; Yu et al., 2016). Therefore, the SWIR band is used to remove some materials that are obviously none water bodies based on experimental thresholds. The collection of new pixels of 'potentially water' after removing non-water materials is denoted as P2, and can be defined as:

$$P2 = \{ i \in P1 \mid \rho_{SWIR,i} < T_{SWIR} \} \tag{5}$$

where pixel i belongs to P1, and $\rho_{SWIR,i}$ is the SWIR band reflectance of pixel i. T_{SWIR} is a specified experience threshold and can be set to be 0.1. This threshold is almost universal considering the calibrated data since it is set to be relatively loose only for removing non-water bodies. The land cover types that are clearly none-water bodies can be removed from potentially water by relying on Eq. (5).

On the other hand, we refine the category 'certainly water' by expanding the spatial distribution of these pixels from 'potentially water' P2 to avoid missing water bodies. Taking into account the impact of the threshold, it is obvious that we can no longer directly use the water index or SWIR band to distinguish water bodies and non-water materials in P2. In this study, we propose the water index roughness (*WIR*) and use the spatial variability of the water index to extract pixels of 'certainly water' in the already assigned 'potentially water' areas, which can be computed as follows:

$$WIR_i = WI_i - \min_{j \in N_i} \{ WI_j \} \tag{6}$$

where N_i is the local neighborhood of pixel i, and a window of size 5 is used to represent the neighborhood in this study. WI_i is the water index value of pixel i. Based on Eq. (6), the *WIR* value of the land cover types of the same category tends to be lower since they have similar water index values. In contrast, the *WIR* value of the water boundary tends to be higher, because the water index values at the water and non-water boundaries tend to be different and the difference in the water index between water bodies and non-water materials is greater than the difference of the same category. The spatial variability of the water index that different spatial locations exhibit different values can be measured by the *WIR*. Thereby, the *WIR* is used to extract more pixels of 'certainly water' at the boundary by setting a threshold, which is combined with

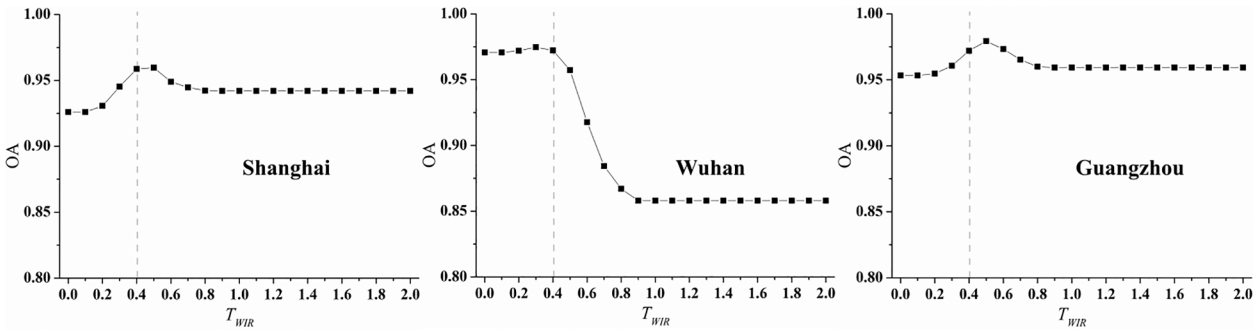


Fig. 6. The sensitivity analysis for the parameter T_{WIR} with the Sentinel-2 MSI data in Wuhan, Shanghai and Guangzhou.

Table 3
Water extraction accuracies using different water indexes.

Study area	Water index	Method	OA (%)	PA (%)	UA (%)	F1
Shanghai	MNDWI	SWM	**95.87**	**93.50**	94.06	**0.9378**
		Optimal threshold	92.20	77.90	**98.36**	0.8694
	AWEI	SWM	**96.27**	**92.20**	**96.44**	**0.9433**
		Optimal threshold	89.00	91.30	78.98	0.8469
	NDWI	SWM	**96.67**	**93.50**	**96.39**	**0.9492**
		Optimal threshold	88.70	74.10	90.26	0.8138
Wuhan	MNDWI	SWM	**97.23**	**93.20**	**98.42**	**0.9574**
		Optimal threshold	88.53	71.70	92.16	0.8065
	AWEI	SWM	**97.63**	**94.10**	**98.74**	**0.9636**
		Optimal threshold	92.27	84.60	91.56	0.8794
	NDWI	SWM	**97.60**	**94.20**	**98.54**	**0.9632**
		Optimal threshold	92.10	80.60	94.94	0.8718
Guangzhou	MNDWI	SWM	**97.20**	**98.40**	**93.54**	**0.9591**
		Optimal threshold	92.53	92.00	86.47	0.8915
	AWEI	SWM	**97.47**	**95.80**	**96.57**	**0.9569**
		Optimal threshold	86.33	93.60	73.01	0.8203
	NDWI	SWM	**96.73**	**98.20**	**92.47**	**0.9525**
		Optimal threshold	84.73	91.80	70.94	0.8003

the already assigned 'certainly water' C1 to obtain new areas of category 'certainly water'. The new pixels of category 'certainly water' is denoted as C2 and can be defined as:

$$C2 = \{\{C1, i \in P2\} | WIR_i > T_{WIR}\} \quad (7)$$

where pixel i belongs to P2, and WIR_i is the water index roughness of pixel i. T_{WIR} is the specified threshold of the water index roughness. In general, considering the difference of the water index between adjacent water bodies and non-water objects, the parameter can be empirically set to be 0.4, and the corresponding sensitivity analysis is discussed in Section 4.1. In the class 'potentially water' P2, if the WIR value of a pixel is greater than the threshold T_{WIR}, it can be classified into 'certainly water'.

2.3. Extracting multi-scale water based on spatial analysis

In the previous section it was described how the categories 'certainly water' and 'potentially water' pixels have been obtained. This section focuses on the methodology to produce the 'final water mask' based on the spatial relationships. The two categories are first of all combined to obtain a segmentation result. The connected-component labeling algorithm is used to perform the segmentation to find regions of spatially connected pixels of the categories 'certainly and potentially water' (Zhao et al., 2015; Zhong et al., 2014). For each segmented region, if there is a spatial connection between both categories, this region may be labeled as surface water. Considering the influence of noise in the water index image, all the pixels in the segmentation region will be considered as surface water to improve the accuracy of water bodies when the number of 'certainly water' pixels meets a certain number or the average water index is greater than the initial threshold T_S. Fig. 3 gives an example of the water extraction based on spatial information. As shown in Fig. 3, the SWM approach can effectively extract water pixels based on the spatial analysis of 'certainly water' and 'potentially water' pixels by eliminating interference from other non-water land cover types.

To quantitatively evaluate the results of the water extraction, we randomly selected evaluation sample points on each experimental data set based on optical false-color images and Google Earth high-resolution images of the corresponding period. Four types of accuracy indicators are used in this study to assess the water extraction results, which include overall accuracy (OA), F1 score (F1), producer's accuracy (PA), and user's accuracy (UA). PA represents the proportion of the pixels which are correctly identified to be water in real reference data of the water body. These can reflect the probability of water being missed (omission error) by 1-PA. UA represents the percentage of real water pixels being classified into the category 'water', which can reflect the probability of incorrect water extractions (commission error) by 1-UA. To comprehensively measure omission error and commission error, F1 score is used, which is defined as a weighted average of PA and UA. OA is the proportion of correctly classified pixels.

3. Experimental areas and data

To evaluate the effectiveness of the proposed SWM algorithm, we selected three representative cities in China (Guangzhou, Shanghai, and Wuhan) as study areas. These cities have different urban characteristics, i.e., a complex pattern of built-up, natural landscapes, and surface waters, which affect the extraction of water bodies. Guangzhou is the capital and most populous city of the Guangdong Province in southern China and has a wide water area with many rivers and water systems located in the Pearl River system. As an important international city, Shanghai is one of the four direct-administered municipalities of China located on the southern estuary of the Yangtze River and has a dense network of rivers composed of the Huangpu River and other tiny river flows inside the city. As the most populous city in Central China, Wuhan is named "city of hundreds of lakes", and has lakes, rivers of different scales, and a large water network consisting mainly of the Yangtze River with a subtropical monsoon climate which is hot and rainy in summer and mild and less rainy in winter. The locations of these study areas can be found in Fig. 4.

The Sentinel-2 multispectral imager (MSI) data are used in this study as these data are free of cost, with global availability and comparatively high spatial resolution. The Sentinel-2 sensors are composed of a two-satellite system including Sentinel-2A and Sentinel-2B. The Sentinel-2 has a high temporal resolution and can achieve a high revisit frequency of about 5 days at the equator and 2–3 days at mid-latitudes, which improves the possibility of obtaining cloudless optical data. The free-of-charge optical data of the Sentinel-2 MSI is currently widely used. They have 13 spectral bands in the visible, near-infrared, and short-wave infrared part of the spectrum with different resolutions from 10 m to 60 m. The spatial resolution of the spectral bands commonly used for water body extraction is concentrated in 10 m (visible and near-infrared bands) and 20 m (short wave infrared bands). These used spectral bands are resampled to 20 m to extract water bodies in this study.

To analyze the performance of the SWM algorithm on time-series optical data with different sensors, we selected Nantan Lake in Huangshi City near Wuhan in the Hubei province as a study area (Fig. 4), because there are many small water bodies with different types and shapes. Besides, Nantan Lake is a key protected lake in the Hubei Provincial Lake Protection List. The lakes in Wuhan and the nearby region show certain seasonal changes with precipitation. When there is sufficient precipitation in July and August, the water area will also increase to some level. Especially when floods occur, the area of the water body will change significantly. We selected the Landsat series optical data of August in 2007, 2011, 2013, 2016, 2019, and 2021 to analyze the yearly changes of this lake. These optical data were acquired by the Landsat Thematic Mapper (TM) (Landsat 5), the Enhanced Thematic Mapper Plus (ETM+) (Landsat 7), and the Operational Land Imager (OLI) (Landsat 8) so that we can verify the effectiveness of the algorithm on different sensors. To validate that Sentinel-2 data can be used to replace

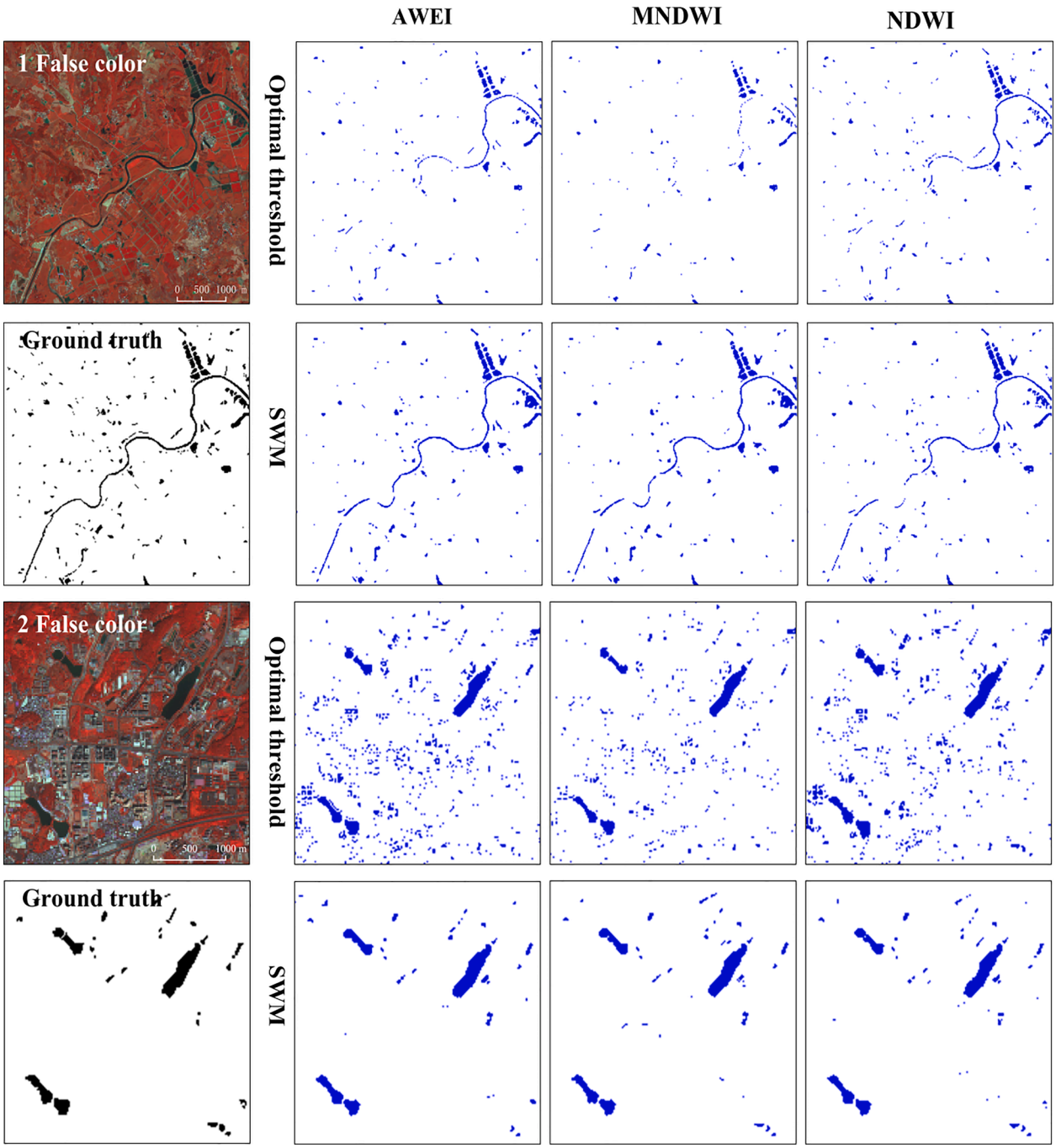


Fig. 7. The water extraction results using the proposed SWM method and the optimal threshold method based on different water indexes at two sample areas from (1) Wuhan and (2) Guangzhou.

missing Landsat data, we also used the Sentinel-2 data with acquisition time close to the existing Landsat data for comparison and production of ground truth. In addition, we also used the Landsat 7 and Landsat 8 data in 2020 to analyze the monthly changes of Nantan Lake. For some months when the Landsat data are absent, the Sentinel-2 data acquired in those months are used as a substitute. A summary of these used images is in Table 1.

4. Results and discussions

4.1. SWM results of a large spatial coverage

We test the performance of the proposed SWM algorithm on the Sentinel-2 MSI data covering the cities of Wuhan, Shanghai, and Guangzhou. Several up-to-date water extraction methods including supervised random forest (RF) (Chen et al., 2020b), hierarchical clustering (HC) (Cordeiro et al. 2021), and multi-band thresholding (MBT) (Chen et al., 2020a) are also performed on the same data as comparisons. Here the SWM algorithm is based on the commonly used water index MNDWI. The Otsu threshold method based on MNDWI is also used for comparison.

The extracted surface water results by the proposed SWM are shown for our three sample cities at the top line of Fig. 5. Six sample areas of small water bodies are selected and enlarged with ground truth data and the water extraction results by all mentioned algorithms are presented in the following lines No. 1–6 (Fig. 5). In addition to the illustration of the

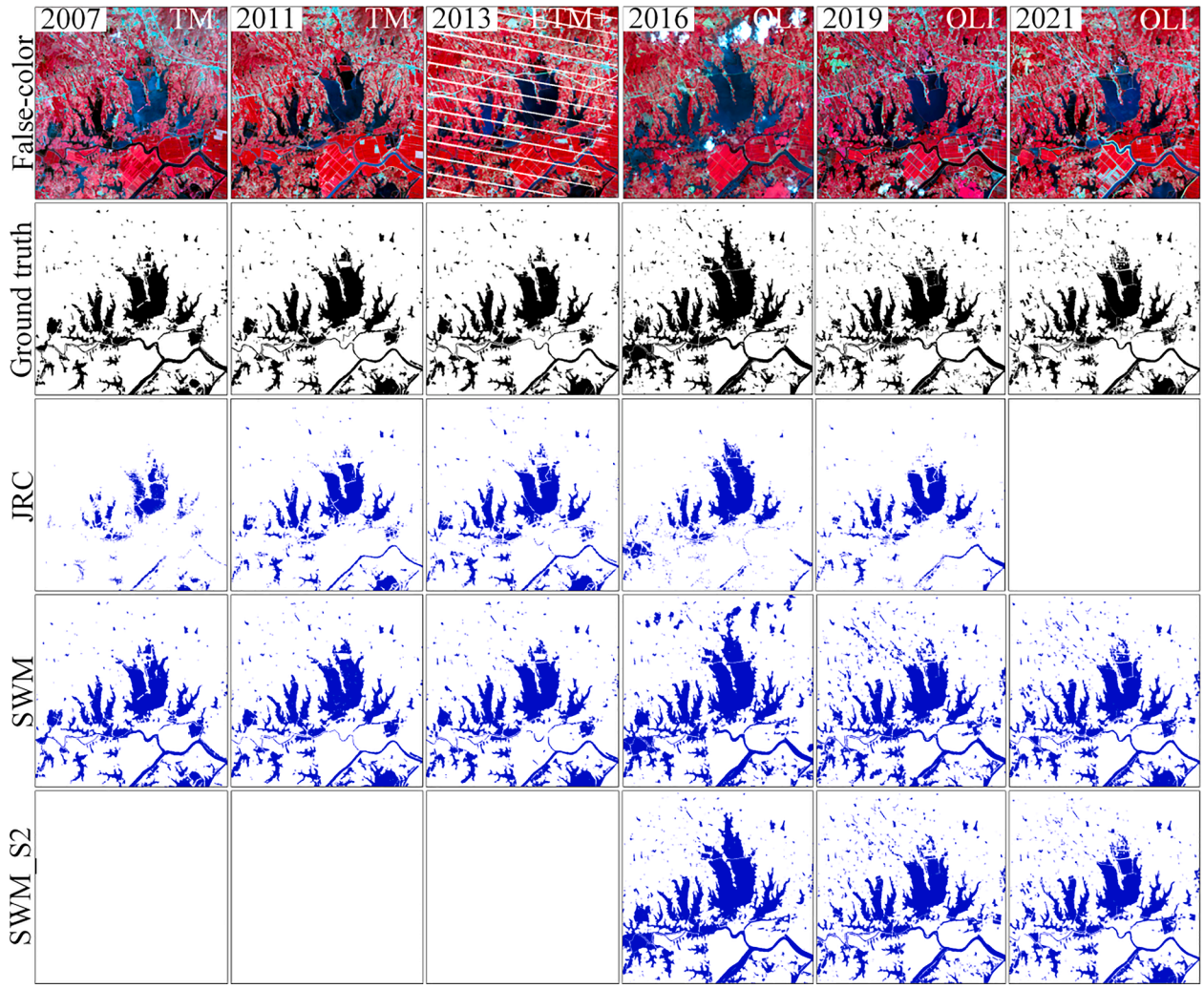


Fig. 8. Inter-year water extraction results at Nantan Lake. 1st line: Landsat false-color images from 2007 to 2021. 2nd line: Ground truth data of the water bodies. 3rd line: JRC product. 4th and 5th lines: water extraction results of the SWM method based on the Landsat and Sentinel-2 images. The Landsat images were acquired by the Thematic Mapper (TM) and the Operational Land Imager (OLI) on 2007/08/09, 2011/08/20, 2013/08/01, 2016/08/01, 2019/07/25 and 2021/07/30. The Sentinel-2 images were acquired by the MultiSpectral Instrument (MSI) on 2016/07/29, 2019/07/29, 2021/08/02.

water extraction results in Fig. 5, the overall accuracy (OA), F1 score (F1), producer's accuracy (PA), and user's accuracy (UA) were used to quantitatively evaluate the accuracy of the algorithm in Table 2.

Generally, all the methods applied for water extraction have achieved acceptable performances, with OA and F1 accuracy indicators greater than 86% and 0.75 respectively. However, these algorithms are show different results for making a balance between reducing incorrect water extractions and increasing the integrity of water extraction. The UA values of Otsu, MBT, and HC methods are 98.36%, 98.81%, and 100% for Shanghai, Wuhan, and Guangzhou, respectively, while the corresponding PA values are 77.9%, 83.1%, and 60.2%. This is mainly because it is a challenge to extract small-scale water bodies in cities for these water surface methods, considering the spectral variability of pixels. Compared with these commonly applied algorithms, the proposed SWM algorithm can extract small water bodies more effectively to reduce the omission error. Among all these methods applied for water extraction, the proposed SWM achieves the highest OA and F1 accuracy for the three experimental areas. Similar conclusions can be drawn from the qualitative results in Fig. 5 at six representative small water bodies. The proposed SWM method proves superior to other algorithms in terms of the integrity of these water bodies especially in small-scale details and with less incorrect extractions, confirming the effectiveness of the proposed algorithm. In comparison to the methods of image classification and clustering e.g., RF and HC, the threshold method based on spectral analysis is easy to be implemented. A single threshold (e.g., Otsu or optimal threshold) is difficult to extract water bodies with fewer omissions and incorrect extractions and is often used as the result of the benchmark. The performance of the MBT method is much improved in comparison to the single threshold method by applying different thresholds to multi-bands, while its performance is still inferior to the proposed SWM method. Overall, the proposed SWM algorithm has a competitive water extraction performance in terms of qualitative and quantitative metrics.

For the SWM algorithm, its threshold parameters have a certain influence on the results of surface water extraction. Most parameters can be automatically selected, such as the high threshold (T_{pure}) and the low threshold (T_{mixed}). Only the threshold of water index roughness (T_{WIR}) needs to be determined, which mainly focuses on how to further refine 'certainly water' from 'potentially water' pixels. Therefore, the sensitivity analysis of the parameter T_{WIR} with the Sentinel-2 MSI data in Wuhan, Shanghai and Guangzhou is given in Fig. 6. Based on Eq. (6), we can find that the range of T_{WIR} is [0–2], thus the threshold T_{WIR} varied between 0 and 2 with an interval of 0.1 to analyze the relationship with the overall accuracy. As shown in Fig. 6, the accuracy of surface water extraction initially increases gradually or remains stable with the increase of the threshold T_{WIR}, since it can gradually exclude some non-

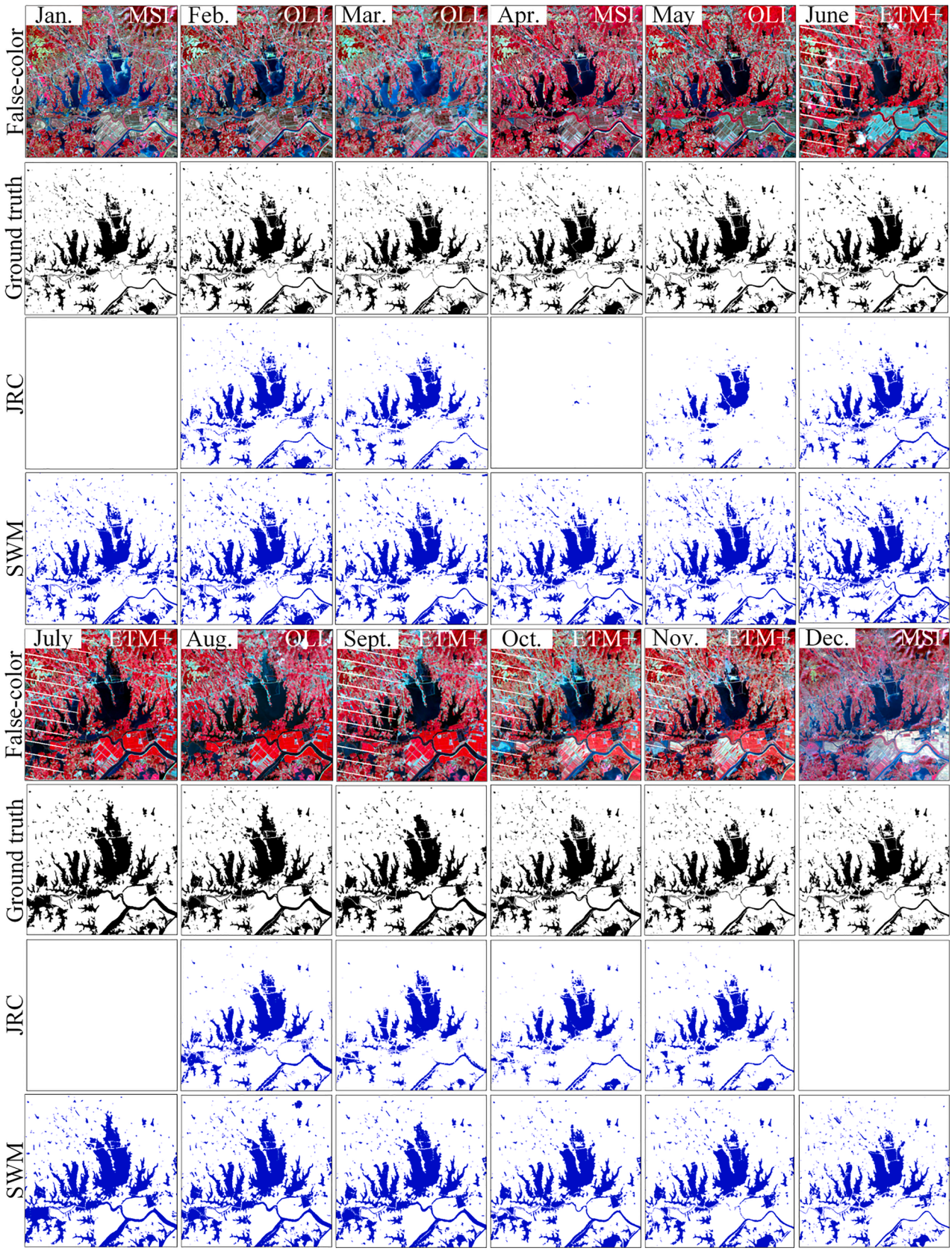


Fig. 9. Monthly water extraction results of the year 2020 at Nantan Lake. Note: there is no applicable optical data acquired in July 2020, so the data acquired by Landsat 7 Enhanced Thematic Mapper Plus (ETM+) at 4 August 2020 is adopted as a substitute.

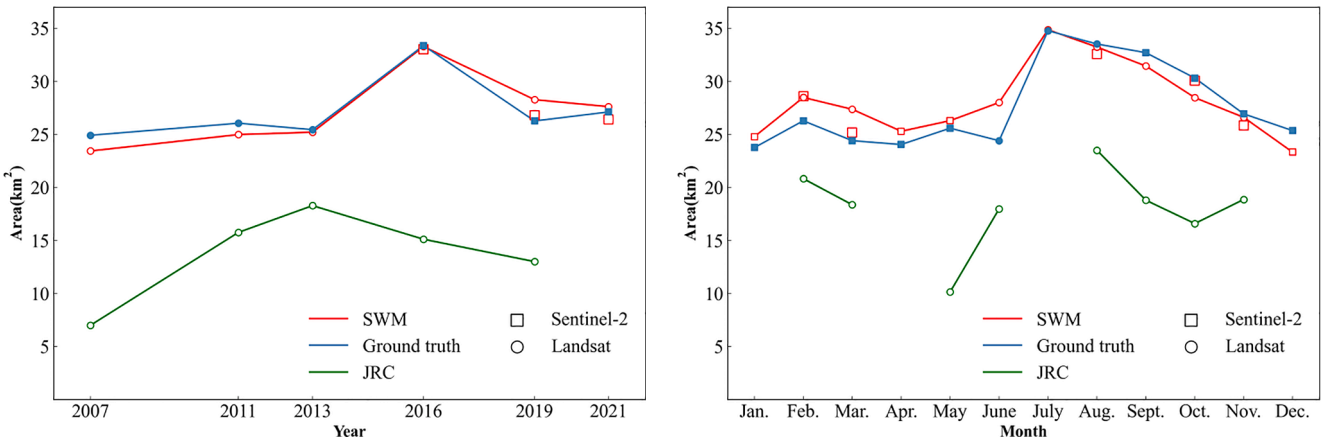


Fig. 10. The inter- and intra-year area extent changes of Nantan Lake extracted by SWM in comparison to ground truth and JRC products.

water objects that are easily confused with the water bodies. With it incorrect water extractions can be reduced when refining 'certainly water' from 'potentially water'. When the threshold reaches a certain value, some small water bodies may have no pixels to be incorporated into the refined 'certainly water' class, resulting in an increasing number of missed small water bodies. Thereby, the accuracy tends to decrease, and then tends to be at a stable state when the threshold is too large, i.e. the water index roughness no longer has its effect. However, we found that a water index roughness threshold of 0.4 is general to some extent, since the parameter is mainly concerned with the difference of the water index between adjacent water bodies and non-water objects. When the threshold is empirically set to 0.4, the proposed SWM algorithm can achieve optimal or near-optimal water extraction results, as shown in Fig. 6.

4.2. SWM results with different water indexes

Different water indexes combine different spectral bands of optical images to enhance the discrimination between water bodies and non-water objects. The proposed SWM algorithm is implemented on the adopted water indexes. By now the most widely used water indexes include the NDWI, MNDWI, and AWEI (Fisher et al. 2016; Ma et al., 2019). In Section 4.1, the SWM algorithm was applied on the MNDWI index. In this section, we analyze the performance of the proposed SWM algorithm using different water indexes including the NDWI, MNDWI, and AWEI to verify the generalization ability of the algorithm.

The same experimental area and data as Section 4.1 were used. These water indexes are used to extract water bodies by a selected optimal threshold between the empirical threshold 0 recommended in many studies (Feyisa et al. 2014) and the automatic threshold obtained by the Otsu method. The water extraction accuracies of the SWM algorithm and the optimal threshold method using these three water indexes are reported in Table 3. Two sample regions from Wuhan and Guangzhou are selected to present the water extraction results based on SWM and optimal threshold methods in Fig. 7.

As can be seen from Table 3, the water indexes are effective techniques to improve the discrimination of water bodies and non-water objects, and already a simple threshold can obtain a good OA for water body extraction. However, reducing incorrect water extractions and increasing the integrity of small surface water extraction at the same time through a single threshold is not possible. In comparison, the proposed method uses different thresholds and integrates spatial information to identify water bodies in different water body index ranges. These water bodies usually have different spatial scales, and we have shown that through spatial analysis classifications are more complete without introducing other non-water objects. With it, the proposed algorithm obtained more than 92% accuracy on both PA and UA indicators. For the comprehensive quantitative accuracy index (OA and F1), the proposed SWM algorithm still has a significant improvement compared to threshold methods for the three widely used water indexes. OA has been improved by more than 3%, and F1 by more than 0.0676. In Fig. 7, the applicably of the SWM method with different water indexes can be qualitatively validated. At the first sample area in lines 1 and 2 of Fig. 7, the optimal threshold fails to completely extract the water body while the SWM method can extract the water bodies more complete based on these different water indexes. On the contrary, at the second sample area in lines 3 and 4 of Fig. 7, the problem of the optimal threshold method is the incorrect extraction of small water bodies, while the SWM method significantly reduces the false extraction, and the performance is stable on different water indexes.

4.3. SWM results of inter- and intra-year dynamics of Nantan Lake

In this section, we analyze the effectiveness of the SWM in extracting water bodies from time-series optical remote sensing images. In addition to the difficulties of complete and accurate small water body extraction, it is also challenging to extract small water bodies from time-series optical images due to the time-series variability of the spectrum. This is mainly caused by changes in surface features and different calibration characteristics of multiple sensors, among other aspects. The threshold-based method is simple and efficient and does not need to solve the problem of collecting training samples, but it is difficult to set a proper threshold. To overcome the restriction of the spectral variability on the threshold method, we propose the SWM method which integrates spatial information and automatically determines the threshold achieving complete and accurate time-series small water body extraction. For the optical images acquired by Landsat 7 ETM+, we applied pre-processing to remove the stripes, thus the extracted water bodies are unaffected by image stripes (Fig. 8).

In Fig. 8, we have selected 6 Landsat images acquired in the years 2007, 2011, 2013, 2016, 2019, and 2021; and 3 Sentinel-2 images acquired in the years 2016, 2019 and 2021. All of these data were obtained in July or August when there is plenty of rain around Wuhan. The Landsat images and the corresponding extracted water bodies are in the first and the fourth rows of Fig. 8. The water extraction results from the Sentinel-2 images are shown in the last row of Fig. 8 and represented as SWM_S2. The ground truth data in the second row of Fig. 8 are generated by expert interpretation and field investigation. Besides, the JRC products are also used for comparison in the third row of Fig. 8. Through the qualitative evaluation, we found that in Fig. 8, the SWM results of the Landsat images are very similar to the ground truth; the details of small water bodies are well extracted in the Landsat SWM results, while the JRC products have a significantly higher rate of omissions. The SWM results of Sentinel-2 are close to the results of the Landsat images

acquired at the same period; both of them extract the water bodies of different years more completely and accurately compared to the JRC products.

It is more difficult to extract the monthly results of water bodies considering the lower availability of data and the influence of clouds. JRC's monthly products are still missing 4 months of results (Jan., Apr., July, and Dec.) in Fig. 9. Therefore, it is necessary to integrate the optical data obtained by different sensors. To extract the monthly water bodies in 2020, the used sensors include Landsat 7 ETM+, Landsat 8 OLI and Sentinel-2 MSI in Fig. 9. The stripes of the ETM+ image have been removed by preprocessing, which does not affect the water extraction results. Through visual analysis of Fig. 9, it can be seen that the results extracted by SWM are evidently continuous and complete, close to the ground truth, and are significantly better than the results of JRC products.

Fig. 10 shows the trend change map of the extracted water area, including the results of SWM extraction based on Landsat (circle) and Sentinel-2 (square) data, the ground truth, and the JRC results. It is illustrated that the results extracted by the SWM are similar to the ground truth, but the results of the JRC have a large deviation. At some time points, the SWM results of Landsat and Sentinel-2 images with temporally similar acquisition time were used to evaluate the stability of the SWM method for remote sensing images acquired by different satellite missions and different sensors. In the case of joint analysis of multiple sensor data, these data can be used for the calibration function. There are three calibration points (2016, 2019, 2021) on the annual chart and five calibration points (Feb., Mar., Aug., Oct., Nov.) on the monthly chart. There is a small deviation between red circles and squares at these calibration points.

For the results in Fig. 10, we also calculated the deviation of the water area of different products, which is used to quantitatively measure the extraction accuracy. The relative deviation value is calculated by the ratio of the deviation value of the extracted water area to the total water area calculated from the ground truth. For the annual results, the mean absolute deviations between the Landsat SWM results, the Sentinels-2 SWM results, the JRC products, and the ground truth are 3.5%, 1.9%, 49.0%, respectively; for monthly results, the numbers are 6.0%, 5.1%, 35.0%, respectively. The mean absolute deviations between the SWM results of Landsat and Sentinel-2 images at the calibration points are 3.6% and 3.9% for annual and monthly results, respectively. Both qualitative and quantitative results verify the effectiveness and accuracy of the proposed SWM method to time-series optical images for automatic extraction of water bodies.

Based on more reliable inter-annual and monthly surface water change results extracted by the SWM, the time-series change characteristics of Nantan Lake can be analyzed. For the annual results, we found that the water area reached its peak in August 2016. This was due to the extreme flooding caused by heavy precipitation in Wuhan and its nearby region in the summer of 2016. The abundant precipitation made the water area increase significantly. From the monthly results, the water area reached its peak in July, and was in a larger state in August and September, and then gradually declined. This is in full compliance with the local characteristics of abundant rainfall in summer.

5. Conclusions

This study aimed at solving two major problems of extracting small water bodies from optical remote sensing images. One is the dilemma that it is difficult to reduce incorrect water extractions and increase the integrity of water extraction at the same time due to the spectral complexity of water and non-water objects. The other one is the spectral variability of the time-series optical images due to the temporal surface change or different calibration accuracy of different sensors. The proposed SWM approach develops a strategy to integrate the spatial information to make up for the indivisibility of spectral information of small water bodies. For the first time, the roughness of the water index is proposed to measure spatial variability of surface water and extract more complete small water bodies considering the large spatial variability of small water bodies and surrounding land cover categories. The proposed SWM method with the automatically determined thresholds has proven to be applicable for different water indexes.

The experimental results using Landsat and Sentinel-2 data from three different cities illustrate the proposed algorithm is capable of achieving consistently better surface water extraction performances, i.e. more complete small-scale water bodies, and it alleviates incorrect water extractions. Compared with the other state-of-the-art water extraction methods, the proposed algorithm can achieve better water extraction performance with significant improvements for overall accuracy (OA and F1). The proposed algorithm can achieve an average OA and F1 score of approximately 97% and 0.95, respectively. The time-series water extraction results of Nantan Lake by the SWM agree well with the ground truth; the water extraction accuracy of SWM is more than 10 times higher than that of JRC products in terms of water body area deviation. These experiments show the proposed SWM method is capable to provide satisfactory small water body extraction results in a long time sequence or short time interval application.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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